

Chapter 1

Embedded Digital Communications B

ENEL712 Embedded Systems Design — Lecture 6

Summary

Beyond simple serial: differential signalling, cable construction, USB topology and stacks, and Ethernet implementation on MCUs.

Topics Covered

- Beyond basic serial protocols
- Single-ended vs differential signalling
- Cable construction and distance
- Universal Serial Bus (USB)
- Ethernet on MCUs

1.1 Beyond Basic Serial Protocols

- SPI/TTL RS-232 typically max ~125 kB/s and rarely sustain that beyond 0.5 m.
- Wireless options (Bluetooth, ZigBee) help with distance but high-bandwidth/long-distance needs require advanced modes.

1.2 Single-Ended vs Differential Signalling

1.2.1 Single-ended

- One signal line + common ground (TTL RS-232, SPI, I²C, PS/2).
- Sensitive to noise, prone to crosstalk, generates EMI.

1.2.2 Differential

- Two complementary lines; receiver subtracts them (USB, Ethernet, HDMI, RS-422/485).
- Common-mode noise cancels at the receiver; reduced EMI; supports higher frequencies.

1.3 Cable Construction and Distance

- Twisted pairs cancel induced EMI and reduce crosstalk; foil shielding blocks external interference.
- Higher-bandwidth cables (Cat.5 → Cat.7a) typically use more shielding.
- Maximum reliable distance is inversely proportional to data rate.
- On long, high-speed lines (incl. USB 3.0), terminating resistors must match the cable's characteristic impedance to absorb signal reflections.

1.4 Universal Serial Bus (USB)

- Host-controlled, tiered-star topology; up to 127 devices across 7 tiers.
- USB 1.0: Low Speed 1.5 Mbit/s, Full Speed 12 Mbit/s. USB 2.0: High Speed 480 Mbit/s. USB 3.0/3.1: SuperSpeed 5/10 Gbit/s.
- Wires: 1.0/2.0 use 4; 3.0 uses 11; USB-C uses 24.
- Enumeration: host detects, addresses, and reads descriptors from a new device.
- Endpoints: unidirectional 'pipes' for Control, Bulk, Interrupt, or Isochronous transfers.

1.4.1 Microcontroller implementation

- AT90USB has a USB peripheral that can act as host or device, with default Endpoint 0 plus 6 programmable endpoints.
- Manufacturers (e.g., Atmel) provide free software stacks (ASF-USB) to handle low-level register and protocol details.

1.5 Ethernet on Microcontrollers

- Most 8/16-bit MCUs lack the RAM and speed for native Ethernet.
- Implementations typically use a dedicated controller (e.g., Microchip ENC28J60) over SPI plus magnetics (transformers) for the RJ-45 socket.
- Writing the TCP/IP stack from scratch is a huge effort; using Linux on ARM or pre-built modules (e.g., XPort) is often more practical.

1.6 Use Cases & Applications

- USB: connect an MCU-based product to PCs without designing a driver — the standard HID class is enough for keyboards, joysticks, and many measurement instruments.
- Ethernet on MCU: factory IoT gateways that aggregate Modbus/CAN sensors and forward them to a server.
- RS-485: multi-drop industrial buses (e.g., 32 sensors on a single twisted pair across a building).
- Differential signalling: HDMI, USB 3+, PCIe, and MIPI in consumer electronics — same idea as RS-422 scaled up to GHz rates.

1.7 Formulas & Rules of Thumb

- Cable bandwidth $\times$ length is roughly constant: doubling the data rate halves the achievable distance.
- RS-485 rule of thumb: 1.2 km at 100 kbps, ~12 m at 10 Mbps.
- Termination: place a resistor matching the cable's characteristic impedance (typically 100–120 Ω) at each end of a long differential bus.
- USB cable length max: 5 m for USB 2.0 High Speed without a hub or active cable; 3 m for USB 3.0.

1.8 Worked Example: Why Long Single-Ended Cables Fail

Imagine running 5 m of unshielded TTL UART (single-ended) past a switching power supply. Each switching pulse couples ~50 mV of noise onto the signal line. With a 3.3 V signal, that 50 mV may not corrupt a steady-state level, but at the edges (where the receiver decides 0 vs 1) it shifts the perceived crossing point and shows up as random framing errors. Switch to RS-485 at the same baud and the receiver sees the noise on both wires equally — common-mode rejection cancels it out.

1.9 Practical Notes

- Don't run differential pairs across split planes — that breaks return-current paths and reintroduces common-mode noise.
- When using an external Ethernet controller (e.g., ENC28J60 + magnetics), keep the SPI bus to the MCU short — SPI is single-ended and noisy.
- USB-C's 24 wires include CC (configuration), power, multiple high-speed pairs, and SBU lines for alternate modes (DisplayPort/Thunderbolt) — most embedded designs only wire the USB 2.0 subset.

Key Definitions

Differential Signalling

Two complementary lines whose difference carries the signal; rejects common-mode noise.

Common-Mode Noise

Interference appearing equally on both lines of a differential pair; cancelled by the receiver.

Twisted Pairs

Conductors physically twisted to cancel induced EMI and reduce crosstalk.

Crosstalk

Coupling between adjacent wires causing interference.

Enumeration

USB host's process of detecting, addressing, and configuring a new device.

Endpoint

Logical buffer/pipe in a USB device for one direction of a particular transfer type.

Magnetics

Transformers in Ethernet ports providing isolation and EMI filtering.

TCP/IP Stack

Set of software protocols managing network communication and data routing.

Software Stack

Manufacturer library handling low-level details of a complex protocol (USB, TCP/IP).

Ethernet Controller

Dedicated IC handling the MAC/PHY layers so an MCU can connect to Ethernet.

RS-485

Differential, multi-point standard supporting long distances (up to 1.2 km).

Common-Mode Rejection Ratio (CMRR)

Receiver's ability to ignore signals appearing equally on both differential lines.

Characteristic Impedance

Effective resistance a long cable presents to a fast edge; mismatched ends cause reflections.

Active Cable

Cable with built-in repeaters/equalisers used to extend high-speed links beyond passive limits.

Sources

- Lecture 6.pdf